

Exp.No 1: Simulation of FDMA in MATLAB

Objective 1:

To simulate an FDMA system by allocating individual frequency channels to multiple users and determine the channel bandwidth assigned to each user. Output Parameter: Channel Bandwidth (kHz)

CODE:

```
clc; clear; close all;

fs=1000; t=0:1/fs:1;

TBW=600; Users=3;

CBW=TBW/Users;

fc=[100 300 500];

m=sin(2*pi*10*t);

s1=m.*cos(2*pi*fc(1)*t);
s2=m.*cos(2*pi*fc(2)*t);
s3=m.*cos(2*pi*fc(3)*t);

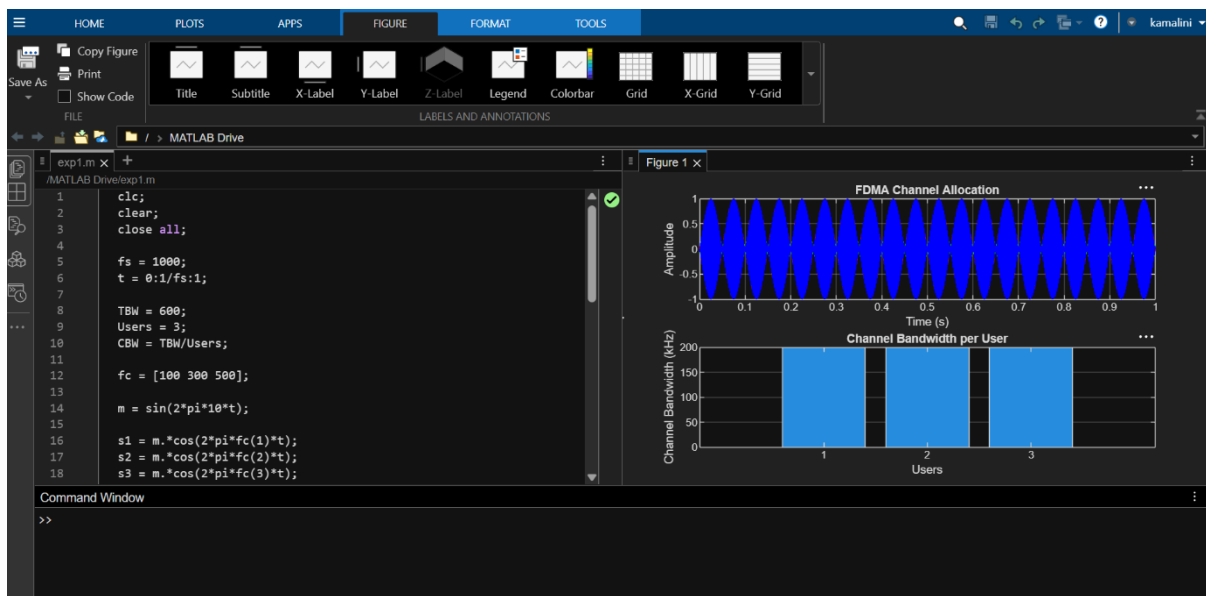
subplot(2,1,1)

plot(t,s1,'r',t,s2,'g',t,s3,'b'),grid on

title('FDMA Channel Allocation'),xlabel('Time
(s)'),ylabel('Amplitude') subplot(2,1,2)

bar(1:Users,CBW*ones(1,Users)),grid on

xlabel('Users'),ylabel('Channel Bandwidth (kHz)'),title('Channel Bandwidth per User')
```



Objective 2:

To calculate the data rate achieved by each user in the FDMA system based on the allocated channel bandwidth. Output Parameter: Data Rate (kbps)

CODE :

```
clc; clear; close all;

BW = 200; % Channel Bandwidth (kHz)

Users = 3;

M = 2; % BPSK Modulation

DataRate = BW*log2(M); % Data Rate (kbps)

Rate = DataRate*ones(1,Users);

disp(' User Data Rate (kbps)')

disp([(1:Users)' Rate'])

figure

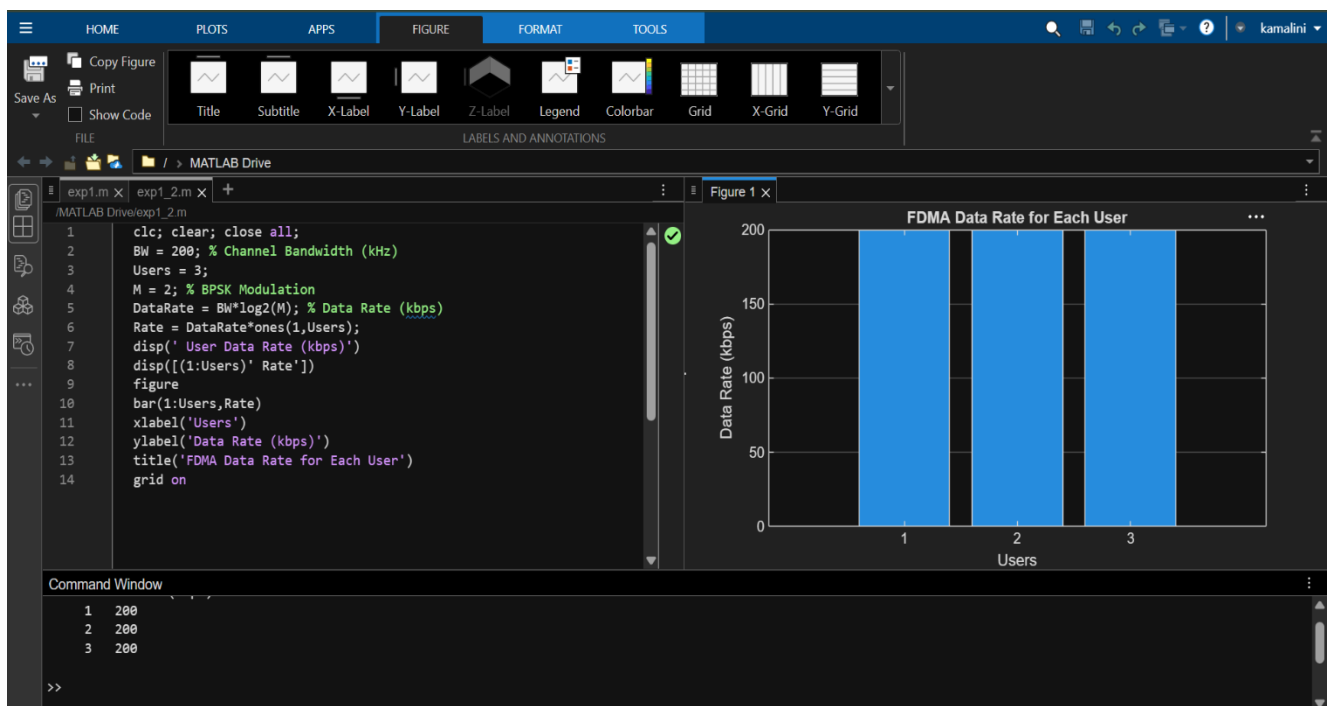
bar(1:Users,Rate)

xlabel('Users')

ylabel('Data Rate (kbps)')

title('FDMA Data Rate for Each User')

grid on
```



Objective 3:

To evaluate the communication quality of the FDMA system by calculating the Signal-to Noise Ratio (SNR) for different users.

CODE :

```
clc; clear; close all;
```

```
Ps = 1; % Signal Power (W)
```

```
Pn = [0.1 0.05 0.01]; % Noise Power (W)
```

```
Users = 1:3;
```

```
SNR = 10*log10(Ps./Pn); % SNR in dB
```

```
disp('User Noise Power(W) SNR(dB)')
```

```
disp([Users' Pn' SNR'])
```

```
figure
```

```
bar(Users,SNR)
```

```
xlabel('Users')
```

```
ylabel('SNR (dB)')
```

```
title('Signal-to-Noise Ratio of FDMA Users')
```

```
grid on
```

